Phase 2 — Solution Requirements

Source-grounded standalone document

Functional Requirements

ID	Requirement
FR-01	Register users through POST /api/auth/register.
FR-02	Authenticate users through POST /api/auth/login.
FR-03	Protect private API endpoints using Bearer JWT authentication.
FR-04	Create FAQ records through POST /api/faqs.
FR-05	Search FAQs through GET /api/faqs/search?q=... .
FR-06	Generate FAQ content through POST /api/ai/generate-faq.
FR-07	Support FAQ CRUD operations.
FR-08	Organize FAQs using categories.
FR-09	Apply Mongoose schema validation before persistence.
FR-10	Provide centralized standardized error handling.

Non-Functional Requirements

ID	Requirement
NFR-01	JWT-based security and bcrypt password encryption.
NFR-02	Data integrity through Mongoose validation.
NFR-03	Layered, modular REST API architecture.
NFR-04	Cross-platform operation on Windows, macOS, and Linux.
NFR-05	Centralized error handling and response sanitization.
NFR-06	Efficient indexing and data discoverability.
NFR-07	Maintainability through modular Express routes and Mongoose ODM.